02 · Lesson spec

Mini-project: Ayo's Robot Dance · 40 minutes (can run over 2 sessions) · teams of 3 · fully unplugged · station check starts

Goal	**I can make Ayo dance a repeating AB pattern and say what comes next.**
Core parts	Code + Build
Key words	Review: pattern · repeat · next · my own
Materials	Cardboard Ayo per team; move cards and blank A–B strips (W8); paper shapes and a card strip for the belt; glue sticks; build card and picture build checklist (Printables); Station 1 kit: pattern strips, hoops and objects, Station 1 record sheet (Guide 05)



1. Launch (5 min)

- **Ask:** "What makes a good robot dance?" Lead to: "A pattern that **repeats**, so friends can copy it and say what comes next!"
- Show the **build card** (6 steps) and the picture build checklist: pattern ✓ repeat 3 times ✓ next move ✓ show ✓.
- Give roles: **card-layer**, **robot** (dances and moves Ayo), **checker** (says "Is it A, B, A, B?").

2. Build (25 min)

- **Build (Core):** decorate cardboard Ayo with a **pattern belt**: glue paper shapes or colours in an AB pattern (circle, square, circle, square…). An adult cuts the shapes.
- **Code (Core):** the card-layer chooses 2 move cards and lays them **3 times in full**: A B A B A B. No ↻ card this week.
- The robot dances the strip while moving Ayo; the checker points to each card. Practise, check, fix, practise again.
- Teams tick their picture build checklist. The teacher **photos** each team with its Ayo and strip.
- **Stretch:** add a third move (A B C) or a sound to the dance.

Meanwhile: Station 1 (teacher, station check starts)

- Take **one small group at a time** (3–4 children, about 4 minutes) to Station 1. Each child: ① copies and continues an AB pattern → ② says what comes next → ③ sorts by one feature and says the rule → ④ makes their own AB pattern.
- Record E / D / S / X on the Station 1 record sheet (Guide 05). Aim for **8–12 children** this period. Continue in Week 12.

3. Show (10 min)

- Each team dances its strip **3 times**, then stops: "What comes next?" The audience answers.
- **Create:** the team presents: "Our pattern is hop, wave… and it repeats!"
- If you can, show another class (walk there together with an adult).

Common mistakes

- Laying A B only once → "Lay it three times: A B, A B, A B."
- Using the ↻ card from Term 1 → "This week we lay the whole pattern so everyone can see it."

- Belt not in a pattern (random shapes) → "Say your belt out loud. Does it repeat?"

What if it doesn't work?

- Not enough time → belt in session 1, dance and show in session 2.
- No cardboard Ayo → a toy, a box with a face, or a child wearing a paper Ayo mask.

Support and stretch

- **Support:** give the team card A; they choose B. The teacher says the pattern with them.
- **Stretch** (optional, never assessed): add a third move (ABC) or a sound.

How we check

Observation, picture build checklist and photo (not marked). Station 1 record sheet for the station check.

> **Safety**
> An adult cuts all shapes and card. Glue sticks only. Space to dance. Walk with an adult to show another class.